

Workshop: One-Way ANOVA and Kruskal-Wallis

your name here

This week, we continue working with the Totally Serious Pop Culture Research Institute™ on research questions involving multiple groups.

Let's call in the packages and data we need,

```
library(tidyverse)
library(ssstats)
binge_watch <- read_csv("https://raw.githubusercontent.com/samanthaseals/STA4173Fa26/refs/heads/main/workshops/data/M2-binge_watch.csv")
franchise_events <- read_csv("https://raw.githubusercontent.com/samanthaseals/STA4173Fa26/refs/heads/main/workshops/data/M2-franchise_events.csv")
training_performance <- read_csv("https://raw.githubusercontent.com/samanthaseals/STA4173Fa26/refs/heads/main/workshops/data/M2-training_performance.csv")
villain_menace <- read_csv("https://raw.githubusercontent.com/samanthaseals/STA4173Fa26/refs/heads/main/workshops/data/M2-villain_menace.csv")
```

Study 1: Villain Archetype Power Levels

The Totally Serious Pop Culture Research Institute conducted an experiment to determine whether different villain archetypes produce different levels of perceived menace. Researchers recruited 90 participants and randomly assigned 30 participants to each of three conditions: The Dark Lord archetype, The Misunderstood Anti-Hero archetype, and The Comically Incompetent archetype.

Participants watched a 2-minute clip featuring a villain of their assigned archetype and then rated the villain's menace on a scale from 0 to 100, with higher scores indicating greater perceived menace. The researchers want to determine whether average menace ratings differ across the three villain archetypes.

Dataset name: villain_menace

Variables:

- archetype
- menace_rating

1a. Find the summary statistics to describe this data.

```
villain_menace |> group_by(archetype) |> mean_median(menace_rating)
```

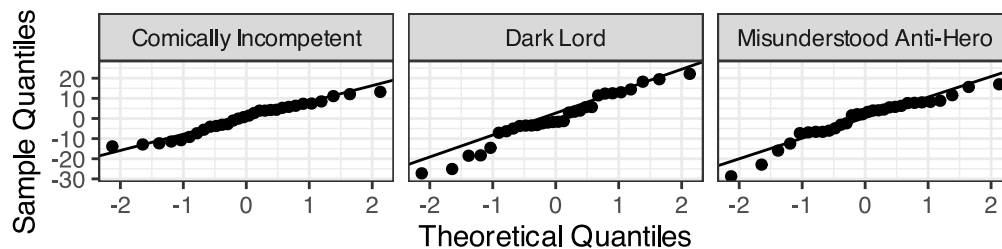
```
# A tibble: 3 × 4
  archetype      variable mean_sd median_iqr
  <chr>          <chr>    <chr>    <chr>
1 Comically Incompetent menace_rating 46.9 (7.8) 48.0 (10.9)
2 Dark Lord      menace_rating 75.7 (12.6) 74.0 (14.7)
3 Misunderstood Anti-Hero menace_rating 66.8 (10.5) 69.5 (13.8)
```

1b. Construct the appropriate q-q plot for this dataset and research question.

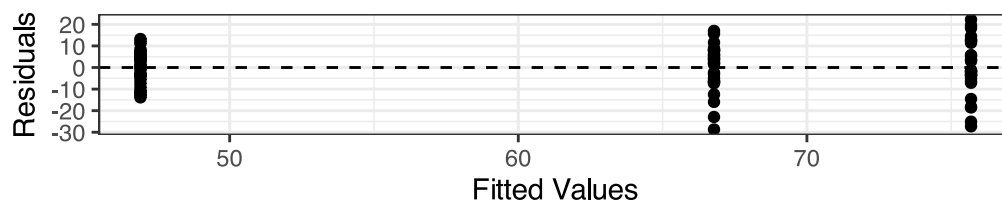
```
villain_menace |> anova_check(outcome = menace_rating,
                              grouping = archetype)
```

One-Way ANOVA Assumptions

QQ Plots of Residuals by Group



Residuals vs Fitted Values



1c. Based on your q-q plot in 1b, do we meet the necessary normality assumption? Why or why not?

Your answer here.

1d. Based on your answers in 1b and 1c, would you test the means or the medians here? Why?

Your answer here.

1e. Construct the appropriate hypothesis test to determine if average menace ratings differ across the three villain archetypes.

```
villain_menace |> anovas(outcome = menace_rating,
                        grouping = archetype)
```

One-way ANOVA

Source	df	SS	MS	F	p-value
archetype	2	13024.2658	6512.1329	59.360	< 0.001
Residuals	87	9544.4069	109.7058		

Hypotheses (archetype):

H_0 : All group means are equal (no effect of archetype)

H_1 : At least one group mean differs (effect of archetype exists)

Test statistic: $F(2, 87) = 59.36$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

```
villain_menace |> kruskal_wallis(outcome = menace_rating,  
                                grouping = archetype)
```

Kruskal-Wallis test

M[Comically Incompetent] = 47.9688 (n = 30)

M[Dark Lord] = 73.9961 (n = 30)

M[Misunderstood Anti-Hero] = 69.4792 (n = 30)

Hypotheses:

H_0 : All group distributions are equal (no effect of archetype)

H_1 : At least one group distribution differs in location (effect of archetype exists)

Test statistic: $H(2) = 51.345$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- *statistic* =
- *p* =

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...

1f. Perform an appropriate posthoc test to identify which villain archetypes differ significantly in menace ratings. Let's treat this as a confirmatory study.

```
villain_menace |> posthoc(outcome = menace_rating,
                          grouping = archetype)
```

Post-hoc analysis: Tukey's HSD

Following: one-way ANOVA (menace_rating ~ archetype)

Comparison	Difference	p-value	
Dark Lord vs Comically Incompetent	28.7782	< 0.001	*
Misunderstood Anti vs Hero vs Comically Incompetent	19.8732	< 0.001	*
Misunderstood Anti vs Hero vs Dark Lord	-8.9050	0.004	*

* $p < \alpha = 0.050$

```
villain_menace |> posthoc(outcome = menace_rating,
                          grouping = archetype,
                          nonparametric = TRUE)
```

Post-hoc analysis: Dunn's test (Bonferroni adjustment)

Following: Kruskal-Wallis (menace_rating ~ archetype)

Comparison	z	p-value	
Comically Incompetent vs Dark Lord	-6.943	< 0.001	*
Comically Incompetent vs Misunderstood Anti-Hero	-5.006	< 0.001	*
Dark Lord vs Misunderstood Anti-Hero	1.937	0.079	

* $p < \alpha = 0.050$

1g. Which groups, if any, are significantly different from one another?

Your answer here.

Study 2: Superhero Training Program Performance

The Totally Serious Pop Culture Research Institute tracks the effectiveness of three different superhero training programs. They recruited 75 participants and randomly assigned 25 participants

to each program: The Classical Hero Academy, The Gritty Realism Program, and The Anime-Inspired Intense Program.

After 12 weeks of training, each participant completed a standardized performance assessment scored from 0 to 100, with higher scores indicating better overall performance. The researchers want to determine whether average performance scores differ across the three training programs.

Dataset name: training_performance

Variables:

- program
- performance_score

2a. Find the summary statistics to describe this data.

```
training_performance |> group_by(program) |> mean_median(performance_score)
```

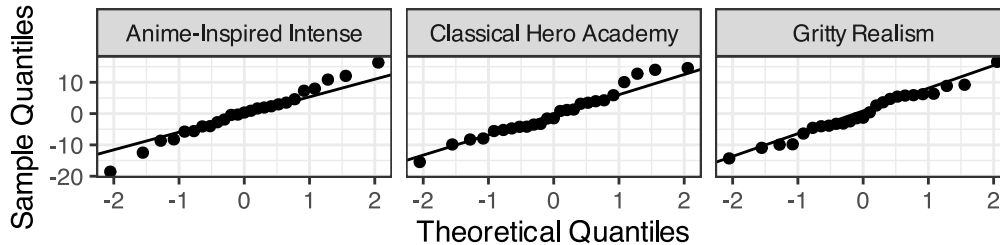
```
# A tibble: 3 × 4
  program          variable    mean_sd    median_iqr
  <chr>           <chr>         <chr>      <chr>
1 Anime-Inspired Intense performance_score 75.1 (7.8) 75.4 (7.6)
2 Classical Hero Academy performance_score 71.7 (7.6) 70.3 (8.7)
3 Gritty Realism    performance_score 78.8 (7.4) 77.5 (9.8)
```

2b. Construct the appropriate q-q plot for this dataset and research question.

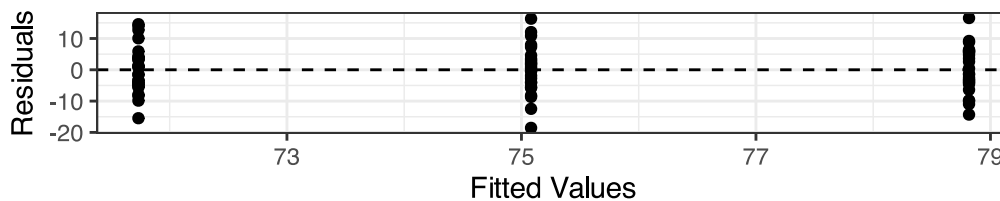
```
training_performance |> anova_check(outcome = performance_score,
                                   grouping = program)
```

One-Way ANOVA Assumptions

QQ Plots of Residuals by Group



Residuals vs Fitted Values



2c. Based on your q-q plot in 2b, explain why Dr. Seals believes we meet the ANOVA assumptions.

Your answer here.

2d. Construct the appropriate hypothesis test (ANOVA) to determine if average performance scores differ across the three training programs.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p -value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...

2e. Perform an appropriate posthoc test to identify which training programs differ significantly in performance. Let's treat this as an exploratory study.

2f. Which groups, if any, are significantly different from one another?

Your answer here.

Study 3: Streaming Service Binge-Watch Behavior

The Totally Serious Pop Culture Research Institute investigated binge-watching habits across subscribers to three different streaming platforms. They recruited 80 participants who actively use one of three services: StreamFlix, PrimateMax, or CinemaPlus.

Over a two-week period, researchers counted the number of episodes each participant watched of their favorite show. The researchers want to determine whether the median number of episodes watched differs across the three streaming platforms.

Dataset name: binge_watch

Variables:

- platform
- episodes_watched

3a. Find the summary statistics to describe this data.

```
binge_watch |> group_by(platform) |> mean_median(episodes_watched)
```

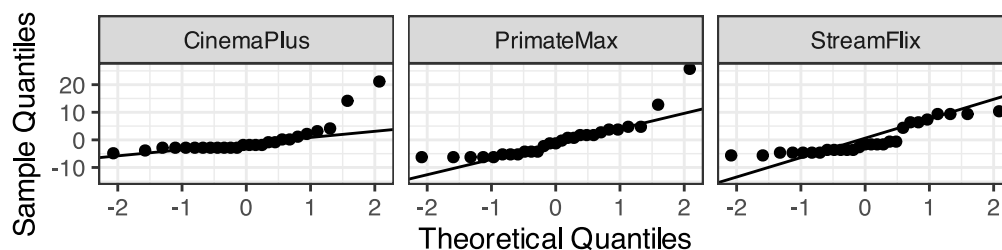
```
# A tibble: 3 × 4
  platform variable    mean_sd median_iqr
  <chr>      <chr>      <chr>    <chr>
1 CinemaPlus episodes_watched 6.8 (5.7) 5.0 (3.0)
2 PrimateMax episodes_watched 8.3 (7.0) 7.0 (7.5)
3 StreamFlix episodes_watched 7.6 (5.5) 6.0 (9.5)
```

3b. Construct the appropriate q-q plot for this dataset and research question.

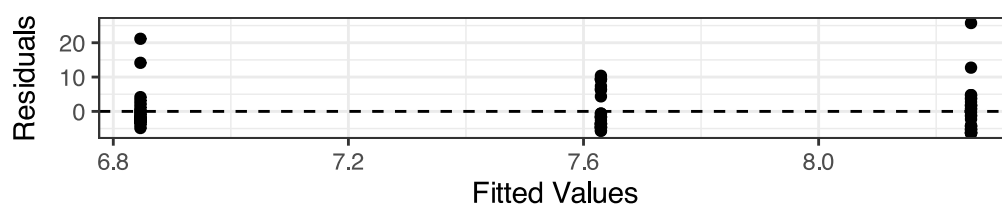
```
binge_watch |> anova_check(outcome = episodes_watched,
                           grouping = platform)
```

One-Way ANOVA Assumptions

QQ Plots of Residuals by Group



Residuals vs Fitted Values



3c. Based on your q-q plot in 3b, do we meet the necessary normality assumption? Why or why not?

Your answer here.

3d. Based on your answers in 3b and 3c, would you test the means or the medians here? Why?

Your answer here.

3e. Construct the appropriate hypothesis test to determine if the number of episodes watched differs across the three streaming platforms.

```
binge_watch |> anovas(outcome = episodes_watched,
                      grouping = platform)
```

One-way ANOVA

Source	df	SS	MS	F	p-value
platform	2	26.5214	13.2607	0.358	0.700
Residuals	77	2852.8661	37.0502		

Hypotheses (platform):

H_0 : All group means are equal (no effect of platform)

H_1 : At least one group mean differs (effect of platform exists)

```
Test statistic: F(2, 77) = 0.358
p-value: 0.700
Conclusion: Fail to reject the null hypothesis (p = 0.700 ≥ α = 0.050)
```

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p -value

- $statistic$ =
- p =

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...

3f. Explain why it is not appropriate to perform posthoc testing in this situation.

Your answer here.

Study 4: Fan Engagement Across Franchises

The Totally Serious Pop Culture Research Institute surveyed fan engagement in four major entertainment franchises. They recruited 100 fans (25 per franchise) from four franchises: The Galaxian Wars, Marvel Legends, The Wizard Chronicles, and Dark Knight Studios.

Researchers counted how many official franchise events each fan attended in the past year. The researchers want to determine whether the median number of events attended differs across the four franchises.

Dataset name: franchise_events

Variables:

- franchise
- events_attended

4a. Find the summary statistics to describe this data.

```
franchise_events |> group_by(franchise) |> mean_median(events_attended)
```

```
# A tibble: 4 × 4
  franchise variable mean_sd median_iqr
```

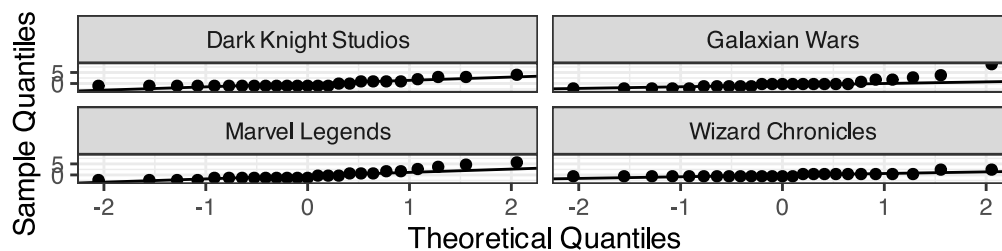
<chr>	<chr>	<chr>	<chr>
1 Dark Knight Studios	events_attended	1.0 (1.5)	0.0 (2.0)
2 Galaxian Wars	events_attended	2.2 (2.4)	2.0 (1.0)
3 Marvel Legends	events_attended	2.3 (2.2)	1.0 (2.0)
4 Wizard Chronicles	events_attended	0.6 (0.9)	0.0 (1.0)

4b. Construct the appropriate q-q plot for this dataset and research question.

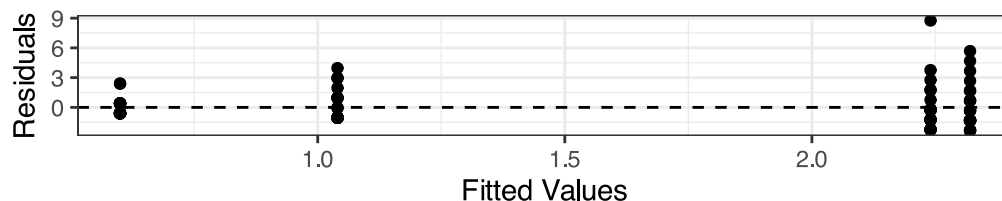
```
franchise_events |> anova_check(outcome = events_attended,
                                grouping = franchise)
```

One-Way ANOVA Assumptions

QQ Plots of Residuals by Group



Residuals vs Fitted Values



4c. Based on your q-q plot in 4b, do we meet the necessary normality assumption? Why or why not?

Your answer here.

4d. Based on your answers in 4b and 4c, would you test the means or the medians here? Why?

Your answer here.

4e. Construct the appropriate hypothesis test to determine if the median number of events attended differs across the four franchises.

```
franchise_events |> anovas(outcome = events_attended,
                           grouping = franchise)
```

One-way ANOVA

Source	df	SS	MS	F	p-value
franchise	3	55.7900	18.5967	5.362	0.002
Residuals	96	332.9600	3.4683		

Hypotheses (franchise):

H_0 : All group means are equal (no effect of franchise)

H_1 : At least one group mean differs (effect of franchise exists)

Test statistic: $F(3, 96) = 5.362$

p-value: 0.002

Conclusion: Reject the null hypothesis ($p = 0.002 < \alpha = 0.050$)

```
franchise_events |> kruskal_wallis(outcome = events_attended,  
                                grouping = franchise)
```

Kruskal-Wallis test

M[Dark Knight Studios] = 0 (n = 25)

M[Galaxian Wars] = 2 (n = 25)

M[Marvel Legends] = 1 (n = 25)

M[Wizard Chronicles] = 0 (n = 25)

Hypotheses:

H_0 : All group distributions are equal (no effect of franchise)

H_1 : At least one group distribution differs in location (effect of franchise exists)

Test statistic: $H(3) = 18.419$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- *statistic* =
- *p* =

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...

4f. Perform an appropriate posthoc test to identify which franchises differ significantly in event attendance. Let's treat this as an exploratory study.

4g. Which groups, if any, are significantly different from one another?

Your answer here.